

Deepali L. Kundnani

Postdoctoral Researcher, Biomedical AI

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EDUCATION

Ph.D. in Bioinformatics

Aug 2019 – Jul 2025

Georgia Institute of Technology · Atlanta, GA

- Minor in Machine Learning · GPA 4.00 / 4.00.
- Thesis: distribution and features of ribomes, the genome-wide embedment of ribonucleotides in eukaryotic genomes.
- Mark Borodovsky Prize for Excellence in Bioinformatics.

M.S. Molecular Biotechnology

Aug 2013 – May 2015

University of Houston – Clear Lake · Houston, TX

- GPA 3.88 / 4.00 · Jones NAS merit scholarship.
- Independent research: yeast two-hybrid screen of an asthma-patient lung cDNA library for iNOS interactions.
- Capstone: high-yield monoclonal antibody production in CHO cells.

B.E. Biotechnology

Aug 2008 – May 2012

Thadomal Shahani Engineering College · University of Mumbai, India

- Third Prize, IEEE national technical paper presentation.
- Thesis: stem cell therapy and wound care for diabetic foot complications.

EXPERIENCE

Postdoctoral Researcher

Apr 2026 – Present

University of New Mexico Comprehensive Cancer Center · Albuquerque, NM

- Developing interpretable AI/ML models integrating EHR, genomics, and multi-omics data for cancer risk prediction.
- Building longitudinal EHR phenotypes, linking clinical variables with large-scale genomic cohorts.
- Contributing to foundation-model research for variant interpretation and tumour genome classification.
- Collaborating across clinical oncology, statistical genetics, and genetic counselling.

AI Strategic Project Advisor, STEM Research

Apr – Jul 2026

Turing · Remote

- Advised a high-priority programme spanning 200+ subject matter experts, tied to \$1M+ revenue.
- Consulted on delivery planning and execution of custom dataset generation for frontier LLM clients.
- Oversaw hiring, onboarding, and weekly targets to hold quality and sign-off timelines.

AI Evaluation Team Lead, STEM Scientific Coding

Dec 2025 – Mar 2026

Turing · Remote

- Led and trained teams of 10–30 SMEs across bioinformatics, computer science, and machine learning.
- Built a cross-domain STEM expert team from scratch across physics, chemistry, maths, biology, and scientific coding.
- Designed rubrics, failure-mode taxonomies, SOPs, QA systems, and SME calibration workflows.
- Contributed to proofs-of-concept, pilots, and large-scale multimodal HLE data generation.
- Built reviewer guidelines, trainer materials, unit-testing protocols, and single-ground-truth factor analysis.

Bioinformatics Subject Matter Expert & Reviewer

Nov 2025

Turing · Remote

- Analysed LLM transcripts to identify model breakage patterns in bioinformatics reasoning.

Graduate Research and Teaching Assistant

Aug 2019 – Jul 2025

Georgia Institute of Technology, Storici Lab · Atlanta, GA

• Storici Lab · Aug 2020 – Jul 2025

- Led research on a novel epigenetic layer in the human genome, analysing multi-omics data to uncover regulatory roles.
- Published novel sequence motifs in a neuro-immune genetic disorder tied to DNA replication and repair.
- Analysed enzyme expression and copy number across CCLE and TCGA in 33+ cancer types, producing an R package for multi-omics integration.
- Contributed the preliminary analysis behind a \$1M grant.
- Conceptualised ribose-seq normalisation and ribonucleotide enriched zones (REZ).
- New data-cleaning protocols for ribose-seq NGS libraries, doubling pattern-recognition accuracy.
- Co-founded the Molecular BioMedical (MBM) Cancer student committee.
- Mentored 4+ early-career bioinformaticians; trained 15+ students in Linux, Python, and R.
- Presented at 8+ national and international conferences.

• Rotational student · Aug 2019 – Jul 2020

- Identified differentially methylated regions in 900+ HIV+ patients (Yale VACS); modelled survival with kNN and SVM.
- Led genome assembly and comparative genomics teams tracing foodborne outbreak sources from CDC data.
- Differential expression and pathway analysis in lung cancer lines across wild-type and multiple p53 mutants.
- Lectured and graded courses for 20–250 students.

Research Assistant II

Oct 2015 – Jul 2019

MD Anderson Cancer Center, Hanash Lab · Houston, TX

- Optimised and validated immunoassays for lung, pancreatic, colon, and ovarian cancer biomarkers.
- Developed auto-antibody assays targeting alternative-splicing products and exosomes.
- Processed patient blood in a CLIA-certified lab for double-blind early lung and breast cancer trials.
- Batch normalisation and missing-data handling for cohorts of 300–7,000+ samples.
- Validated novel protein markers in epithelial–mesenchymal transition (EMT).
- Managed large-scale DNA vaccine production and in-house antigen/antibody production.
- Integrated genomic and transcriptomic data with in-house mass spectrometry.

Laboratory Intern

Jan – May 2015

AnshLabs · Houston, TX

- Led production of highly sensitive antigens in mammalian cell lines under GMP guidelines; supported ISO and FDA audits.
- Screened, purified, and QC'd monoclonal antibodies from hybridoma clones for commercial ovarian cancer ELISA kits.

Graduate Teaching Assistant

Sep – Dec 2014

University of Houston – Clear Lake · Houston, TX

- Liaison between students and faculty on applied biotechnology projects and course queries.
- Trained teaching assistants to set up basic and advanced molecular biology labs.

Research and Teaching Assistant

Sep 2012 – Jul 2013

Usha Biotech Ltd. · Hyderabad, India

- Tested proprietary vectors including the patented CELL EXPRESS-100™, using an eGFP reporter for stable antibody expression in CHO-K1 cells.
- Molecular cloning of custom plasmid vectors and transfection optimisation supporting a patent filing.
- Selected transfected pools and clones by fluorescence-activated cell sorting.
- Trained students in mammalian cell culture.

Trainee

May 2011 – Apr 2012

S. L. Raheja Hospital (a Fortis Associate) · Mumbai, India

- Reviewed literature and monitored clinical study and toxicology reports on stem cell therapy for diabetic foot complications.
- Screened patients against inclusion criteria and supported biochemical testing.
- Developed statistical standards for treatment efficacy from wound-closure measurements.

PUBLICATIONS

- 2026** **Kundnani, D. L.**, Lee, Y., Yang, T., Warner, T. P., Martinez-Figueroa, F., Channagiri, T., Xu, P., ... & Storici, F. *Human nuclear ribosomes reveal DNA-embedded ribonucleotides as epigenetic modulators*. **Cell**
- 2026** Niu, X., **Kundnani, D. L.**, Dicome, M., Tafoya, L., Song, L., Mamedov, M. R., Liu, X. S. & Sahu, A. *Tumor $\gamma\delta$ T-cell abundance is associated with favorable cancer treatment outcomes*. **bioRxiv (preprint)**
- 2025** Moalic, Y., Reveil, M., **Kundnani, D. L.**, Balachander, S., Yang, T., ... Storici, F. & Henneke, G. *Genome-wide ribonucleotide detection in Archaea*. **Nucleic Acids Research**
- 2025** Bazzani, V., **Kundnani, D. L.**, Redin, M. E., Agostini, F., Corso, A., ... Storici, F. & Vascotto, C. *Characterization of a new mutation of the mitochondrial ND6 gene in hepatocellular carcinoma*. **Scientific Reports**

- 2024** Kundnani, D. L., Yang, T., Gombolay, A., Mukherjee, K., Newnam, G., Meers, C., ... & Storici, F. *Distinct features of ribonucleotides within genomic DNA in Aicardi-Goutières syndrome ortholog mutants of S. cerevisiae.* **iScience**
- 2024** Yang, J., Sun, M., Ran, Z., Yang, T., Kundnani, D. L., Storici, F. & Xu, P. *rNMPID: a database for riboNucleoside MonoPhosphates in DNA.* **Bioinformatics Advances**
- 2023** Xu, P., Yang, T., Kundnani, D. L., Sun, M., Marsili, S., Gombolay, A. L., ... Storici, F. *Light-strand bias and enriched zones of embedded ribonucleotides in the human mitochondrial genome.* **Nucleic Acids Research**
- 2021** Kundnani, D. & Storici, F. *FeatureCorr: an R package to study feature correlations for sequencing and microarray data.* **Software Impacts**
- 2021** Marsili, S., Tichon, A., Kundnani, D. & Storici, F. *Gene co-expression analysis of human RNASEH2A reveals networks in DNA replication, damage response, and cell cycle.* **Biology**
- 2020** Ostrin, E. J., Bantis, L. E., Wilson, D. O., Patel, N., Wang, R., Kundnani, D., ... & Hanash, S. M. *Contribution of a blood-based protein biomarker panel to the classification of indeterminate pulmonary nodules.* **Journal of Thoracic Oncology**
- 2020** Subbalakshmi, A. R., Kundnani, D., Biswas, K., Ghosh, A., Hanash, S. M., Tripathi, S. C. & Jolly, M. K. *NFATc acts as a non-canonical phenotypic stability factor for a hybrid epithelial/mesenchymal phenotype.* **Frontiers in Oncology**
- 2020** Capello, M., Fahrman, J. F., Rios Perez, M. V., Vykoukal, J. V., Irajizad, E., ... & Hanash, S. M. *CES2 expression in pancreatic adenocarcinoma is predictive of response to irinotecan and is associated with type 2 diabetes.* **JCO Precision Oncology**
- 2020** Kobayashi, M., Katayama, H., Irajizad, E., Vykoukal, J. V., Fahrman, J. F., Kundnani, D. L., ... & Hanash, S. M. *Proteome profiling uncovers an autoimmune response signature reflecting ovarian cancer pathogenesis.* **Cancers**
- 2019** Capello, M., Vykoukal, J. V., Katayama, H., Bantis, L. E., Wang, H., Kundnani, D. L., ... & Hanash, S. M. *Exosomes harbor B cell targets in pancreatic adenocarcinoma and exert decoy function against complement-mediated cytotoxicity.* **Nature Communications**
- 2018** Jia, D., George, J. T., Tripathi, S. C., Kundnani, D. L., Lu, M., Hanash, S. M., ... & Levine, H. *Testing the gene expression classification of the EMT spectrum.* **Physical Biology**
- 2018** Guida, F., Sun, N., Bantis, L. E., Muller, D. C., Li, P., Taguchi, A., ... & Hanash, S. M. *Assessment of lung cancer risk on the basis of a biomarker panel of circulating proteins.* **JAMA Oncology**
- 2018** Fahrman, J. F., Bantis, L. E., Capello, M., Scelo, G., Dennison, J. B., ... & Hanash, S. M. *A plasma-derived protein-metabolite multiplexed panel for early-stage pancreatic cancer.* **JNCI**

- 2016** Capello, M., Bantis, L. E., Scelo, G., Zhao, Y., Li, P., Dhillon, D. S., ... & Hanash, S. M. *Sequential validation of blood-based protein biomarker candidates for early-stage pancreatic cancer.* **JNCI**

CONFERENCE PRESENTATIONS

- 2024** **Aicardi-Goutières syndrome: an RNase H2C mutation mimics the ribonucleotide patterns of an RNase H2B PIP-box mutation.** Georgia Tech School of Biological Sciences retreat. Full funding
- 2024** **Aicardi-Goutières syndrome: RNase H2C mutation mimics RNase H2B PIP-box patterns.** FASEB Conference, Florida. Travel funding
- 2024** **Aicardi-Goutières syndrome: RNase H2C mutation mimics RNase H2B PIP-box patterns.** MCBIOS Conference, Georgia. Full funding
- 2024** **Aicardi-Goutières syndrome: RNase H2C mutation mimics RNase H2B PIP-box patterns.** Bio-IT World Conference & Expo, Boston. Funded by Combinatics Inc.
- 2024** **CellKb: cell-type markers for annotation and discovery in scRNA-seq datasets.** Bio-IT World Conference & Expo, Boston. For Combinatics Inc.
- 2023** **CellKb: cell-type markers for annotation and discovery in scRNA-seq datasets.** Bio-IT World Conference & Expo, Boston. For Combinatics Inc.
- 2023** **Ribonucleotide incorporation in AGS-orthologous mutants of yeast RNase H2 subunits.** Bio-IT World Conference & Expo, Boston. Funded by Combinatics Inc.
- 2022** **Effect of AGS-orthologous mutations on ribonucleotide incorporation in *S. cerevisiae*.** RNASEH Conference, Maryland
- 2022** **Expression correlation of RNASEH2A in cancer datasets.** Global Virtual Congress on Cancer Research & Drug Development
- 2021** **Expression correlation and copy number alteration prevalence of human RNASEH2A in cancer.** 26th Annual Meeting of the RNA Society. NSF Conference Award
- 2011** **Genetic algorithms.** IEEE-EMBS National Technical Paper Presentation, Mumbai. Third Prize
- 2009** **Water purification systems.** BIOZEAL, Mumbai

INVITED TALKS AND LECTURES

- 2025** **Bioinformatics, Ribose-Map, and the PACE cloud computing server.** Research Project Laboratory (BIOL-4590), Georgia Tech
- 2024** **Distribution of ribonucleotides embedded in the nuclear genome of multiple human cell types.** MBM Seminar Series, Georgia Tech

- 2024** **Distinct features of ribonucleotides in AGS-ortholog mutants in yeast.** MBM Seminar Series, Georgia Tech
- 2023** **The role of RNA in DNA repair and modification.** RNA Biology and Biotechnology (BIOL-4560/8560), Georgia Tech
- 2023** **Bioinformatics and an introduction to the Ribose-Map toolkit.** Research Project Laboratory (BIOL-4590), Georgia Tech
- 2020** **Data preprocessing and dimensionality reduction in biomedical and clinical settings.** Biostatistics (APPH-6225), Georgia Tech
- 2019** **DNA sequencing and phylogenetic analysis.** Genetics Lab (BIOL-2345), Georgia Tech
- 2010** **Time management.** Management course, University of Mumbai

AWARDS

- 2024** Student Government Association Conference Award. *Georgia Institute of Technology*
- 2023–24** Bio-IT World Conference Registration Funds. *Combinatics Inc.*
- 2023** Mark Borodovsky Prize for Excellence in Bioinformatics. *Georgia Institute of Technology*
- 2021** US National Science Foundation Conference Award. *26th Annual Meeting of the RNA Society*
- 2019** Performance Excellence Award. *UT MD Anderson Cancer Center*
- 2014–15** Jones NAS Biological Sciences Scholarship. *University of Houston – Clear Lake*
- 2010** Third Prize, National Technical Paper Presentation. *IEEE, India*
- 2008** Top Student Honour, National Standard Examination. *Top 10% nationally, India*

LEADERSHIP AND SERVICE

- 2022–25** **SKY at Georgia Tech: President, then Advisor** Raised >\$20,000; well-being programmes reaching 1,000+ students.
- 2021–24** **Research mentor, 5 B.S./M.S. bioinformatics students** Now at Mayo Clinic, Penn State, Moderna, and UT Southwestern.
- 2022–24** **Founding committee member, MBM Cancer Talk Series** Georgia Tech's molecular biomedical seminar series.
- 2021** **Grant writing, Storici Lab** Contributed to a \$1M award from the W. M. Keck Foundation.
- 2023** **Emerging Leaders Advisory Board** Advised Georgia Tech's VP of Student Engagement.
- 2020–22** **Asha for Education, Atlanta: Secretary** Ran a half marathon raising funds for children's education in India.

SKILLS

Languages & tools	Python, R, SQL, C++, Java, Bash, Git, Snakemake
AI & machine learning	PyTorch, Interpretable ML, Foundation models, LLM evaluation, Variational autoencoders, scanpy, Survival analysis (kNN, SVM), Deep learning
Sequencing analysis	Next-generation sequencing, Long-read sequencing, ribose-seq, Single-cell RNA-seq, Genome assembly, Comparative genomics
Data & infrastructure	Multi-omics integration, EHR phenotyping, High-performance computing, Cloud computing (AWS), Batch normalisation, Data visualisation
Laboratory	Multiplex immunoassays, Molecular cloning, Mammalian cell culture, ELISA development, FACS, CLIA/FDA-compliant SOPs
Certifications	AWS Generative AI (2025), SQL for Data Science, CLIA, MD Anderson (2018), R Programming, Coursera (2017), Data Scientist Toolbox (2016), ISO 9001:2008 (2015)

